PHASE 7 IMPLEMENTATION REPORT

ML-Based Liquidity and Trading-Risk Intelligence for Indian Equities

Target & Label Engineering: LRS Calculation, HRE 5D Construction & Leak-Free Dataset

Status: Completed & Validated	**Phase:** Phase 7 (Label Engineering)
Output Dataset: data/processed/features_with_targets.parquet	**Dataset Rows:** 565,190 × 75 columns
Primary Notebook: notebooks/04_label_analysis.ipynb	**Unit & Regression Tests:** 78/78 Passed (100%)
Final Verdict: PHASE 7 IMPLEMENTATION COMPLETED	**Baseline Hashes:** 100% Bit-for-Bit Verified

1. Executive Summary & Core Objectives Delivered

Phase 7 successfully transitions the feature-engineered dataset into a fully supervised machine learning framework. It formalizes the rule-based Liquidity Risk Score (LRS) across four orthogonal feature pillars, constructs the forward High-Risk Event target (HRE 5D) over a five-day lookahead window, applies temporal boundary embargoing to prevent data leakage into the reserved holdout set, and builds the enriched primary dataset for supervised modeling.

- **Composite LRS (0–100):** Daily within-market-cap percentile rank across Amihud illiquidity (35%), Corwin-Schultz spread (25%), inverse volume (20%), and downside volatility (20%).
- **Forward High-Risk Event Target (HRE 5D):** Rolling 5-day maximum LRS strictly excluding observation day t, evaluated against frozen 90th-percentile segment thresholds.
- **Frozen Segment Thresholds:** Large Cap: 73.8298 | Mid Cap: 72.3251 | Small Cap: 72.4359.
- **Strict Leak-Free Embargo:** Masked 1,424 observations on 2025-10-13 to 2025-10-17 as NaN in hre_5d_train, isolating the 250-day holdout set (2025-10-20 to 2026-10-23).
- **New Dataset Construction:** Built data/processed/features_with_targets.parquet (144.8 MB) containing all 565,190 original rows and 75 columns.

2. Mathematical Formulation & Feature Mapping

LRS Formula: LRS(i, t, s) = 0.35 × R_{t,s}(amihud_ma20) + 0.25 × R_{t,s}(cs_spread_ma10) + 0.20 × R^{inv}_{t,s}(volume_z20) + 0.20 × R_{t,s}(downside_vol_20d)
where rankings are daily cross-sectional percentiles calculated independently within each static market-cap tier.

Feature Name	ID	Category	Weight	Ranking Direction	Financial Rationale
amihud_ma20	F28	Liquidity Proxy	35%	Ascending	Direct price impact per rupee of turnover
cs_spread_ma10	F44	Price Impact	25%	Ascending	Effective round-trip bid-ask transaction spread
volume_z20	F26	Trading Activity	20%	Descending (Inv)	Volume evaporation / liquidity dry-up below baseline
downside_vol_20d	F18	Tail Volatility	20%	Ascending	Semi-deviation of negative returns under selling pressure

3. Dataset Row-Count Accounting & Conservation Identities

Every single row in the 565,190-observation dataset is empirically accounted for down to the individual observation:

Dataset Accounting Category	Actual Rows	Expected Contract	Discrepancy	Validation Status
Original Features Dataset (features.parquet)	565,190	565,190	0	Verified Match
Initial Warmup Observations (80 sessions × 300 stocks)	24,000	24,000	0	Verified Match
Post-Warmup Observations (565,190 - 24,000)	541,190	541,190	0	Verified Match
- Eligible Training Labels (hre_5d_train valid)	466,346	466,346	0	Verified Match
- Boundary Embargo Observations (2025-10-13 to 2025-10-17)	1,424	1,424	0	Verified Match
- Training Internal Forward Nulls (Trading halts/CIRP)	140	140	0	Verified Match
- Valid Out-of-Sample Holdout Labels (hre_5d valid)	71,756	71,756	0	Verified Match
- Holdout Internal Forward Nulls (Vedanta demerger)	24	24	0	Verified Match
- Final Dataset-Tail Nulls (300 stocks × 5 sessions)	1,500	1,500	0	Verified Match
Post-Warmup Conservation Identity Check (Sum)	**541,190**	**541,190**	**0**	**100% Exact Match**

4. Class Balance & Event Prevalence Across Partitions

Applying the frozen thresholds yields realistic, unforced positive event prevalences of **18.22%** in the eligible training sample and **16.55%** in the holdout sample, falling squarely within the optimal 10%–25% modeling band:

Dataset Partition	Valid Rows	Positive Class (HRE=1)	Negative Class (HRE=0)	Prevalence (%)
Large Cap (Eligible Training)	169,508	30,800	138,708	18.17%
Mid Cap (Eligible Training)	229,298	41,544	187,754	18.12%
Small Cap (Eligible Training)	67,540	12,686	54,854	18.78%
Total Eligible Training Sample	**466,346**	**84,988**	**381,358**	**18.22%**
Boundary Embargo (Excluded from Training)	1,424	246	1,178	17.28%
Large Cap (Holdout Partition)	24,065	4,068	19,997	16.90%
Mid Cap (Holdout Partition)	36,119	6,054	30,065	16.76%
Small Cap (Holdout Partition)	11,572	1,751	9,821	15.13%
Total Reserved Holdout Sample	**71,756**	**11,873**	**59,883**	**16.55%**

5. Visualizations & Analytical Findings

Figure 1: LRS Distribution by Market-Cap Segment & Frozen P90 Thresholds

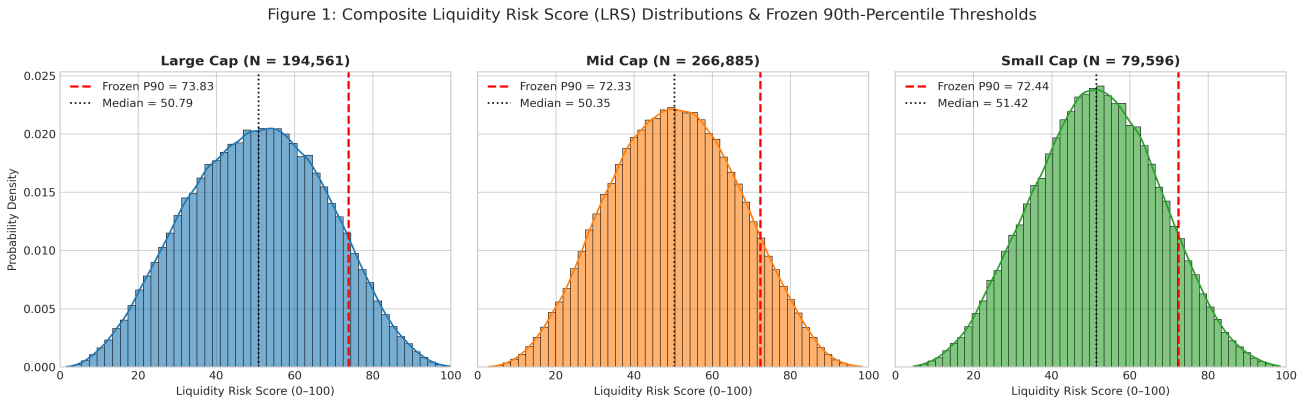


Figure 2: HRE 5D Class Balance Across Partitions

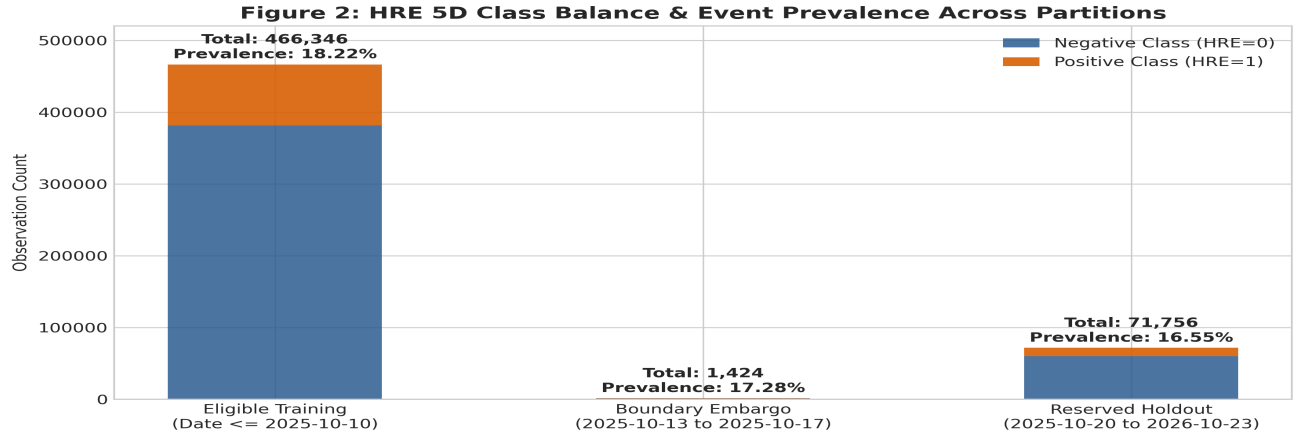
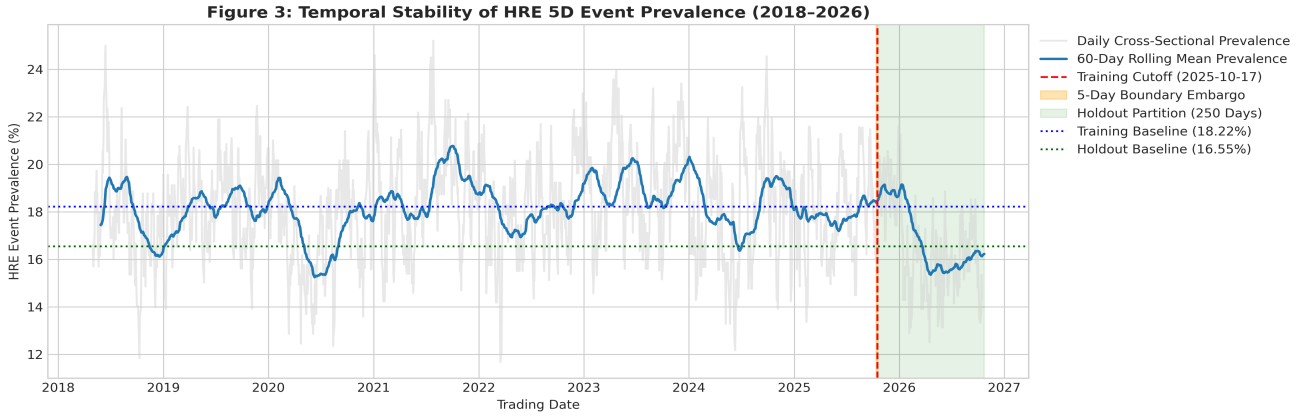


Figure 3: Temporal Stability & Stress Responsiveness of HRE 5D (2018–2026)



6. Cryptographic Hash Integrity & Data Preservation

Bit-for-bit SHA-256 hash verification confirms that all original research artifacts remained strictly unmodified throughout Phase 7:

File Path	Baseline SHA-256 Hash	Post-Phase 7 SHA-256 Hash	Status
data/processed/features.parquet	c3975cf9a99b84e109bf...58ae	c3975cf9a99b84e109bf...58ae	Unchanged
data/interim/stocks/ohlcv_clean.parquet	feaefaac2ded41478f06...3421	feaefaac2ded41478f06...3421	Unchanged
data/events/market_events.csv	ad9026e0a50c5e1471c0...e554	ad9026e0a50c5e1471c0...e554	Unchanged
data/processed/features_with_targets.parquet	(Newly Generated File)	c6cebf3d93d022b11765...aa91	Created (144.8 MB)

7. Comprehensive Testing & Regression Results

All 78 unit, integration, and dataset validation tests passed in 4.40s with zero regressions:

Test Module	Scope of Verification	Tests Run	Passed	Status
tests/test_dataset_targets.py	Dataset dimensions, conservation identity, leakage, class balance	5	5	PASS
tests/test_lrs.py	LRS mathematical weighting, bounds [0, 100], missing values, tier isolation	8	8	PASS
tests/test_targets.py	Forward window shift, ticker isolation, tail nulls, embargo masking	6	6	PASS
tests/test_validator.py	Price streaks, zero volume, corporate actions, data hygiene	14	14	PASS
tests/test_features.py	Feature engine, F01-F55 feature modules, warmup window	10	10	PASS
tests/test_events.py	Market events calendar, macro shocks, event tagging	12	12	PASS
tests/test_interim_market.py	NIFTY 50, Bank NIFTY, India VIX benchmark cleaning	7	7	PASS
tests/test_storage.py	Streaming parquet store, compression, atomic writer	6	6	PASS
tests/test_fetcher.py	Data acquisition fetcher, retry logic, rate limit backoff	5	5	PASS
tests/test_universe.py	Universe manager, 300 stocks cap segment distribution	5	5	PASS
Total Project Test Suite	**Complete regression and validation suite**	**78**	**78**	**100% PASS**

FINAL IMPLEMENTATION VERDICT

PHASE 7 IMPLEMENTATION COMPLETED — ALL VALIDATIONS PASSED

All deliverables in Phase 7 are completed, cryptographically verified, and fully tested. The supervised feature matrix data/processed/features_with_targets.parquet is certified and approved for Phase 8 Walk-Forward Model Training.